USER MANUAL · DESKTOP INTERFACE

# pICNIK Edition Desktop

*Fast, simple, and effective isoconversional kinetic analysis tool.*

Software: pICNIK Edition Desktop • Computation: Isoconversional Kinetics • Language: English

## Contents



## What is pICNIK Edition Desktop?

**pICNIK Edition Desktop** is a specialized graphical desktop application designed to perform non-isothermal thermogravimetric (TGA) isoconversional kinetic calculations quickly, easily, and effectively. It provides an intuitive workflow guiding users through raw experimental data loading, range selection, conversion analysis, activation energy determination, compensation effect fitting, and model reconstruction.

## Before You Start

**File Format.** Plain text files (`.txt`) or comma-separated value files (`.csv`).

**Navigation & Validation.** Users can navigate freely between screens using `Next »` and `« Previous`. If an action is skipped, a deferred validation system halts processing when calculations are requested and prompts the user to complete the missing step.

**Interface Customization.** Theme color and visual styles of *title bar* adapt automatically according to the host operating system. The style of the remaining application components is modified using the options available under the `theme` in the *menu bar*.

# 1 Interface Overview & Navigation

The main workspace is divided into three functional areas:

**Menu Bar:**

- **Files:** Contains `Open files` to load experimental data and `Exit` to close the application.
- **Images:** Contains `Save image` to export the current visualization rendered in the bottom frame.
- **Help:** Contains `About` (link to pICNIK) and user documentation/tutorials.

**Top Frame:** Dedicated exclusively to primary navigation and action buttons (e.g., `Next »`, `« Previous` and `Open Files`).

**Bottom Frame:** Displays the welcome message upon launch and serves as the primary view for rendering plots, charts, and tabular reports.

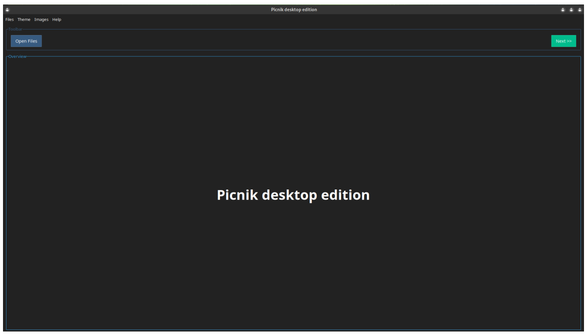



Figure 1: Start screen of the pICNIK desktop GUI, displaying the initial options to load experimental data files via the `Open Files` button.

# 2 Step-by-Step Workflow

## 2.1 Data Loading and Initial Inspection

*Load experimental thermogravimetric data files and perform initial inspection.*

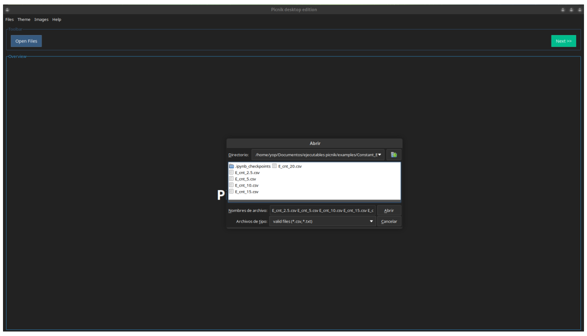
**WHAT YOU NEED**

- Experimental data files (`.csv` or `.txt`) for multiple heating rates $\beta$

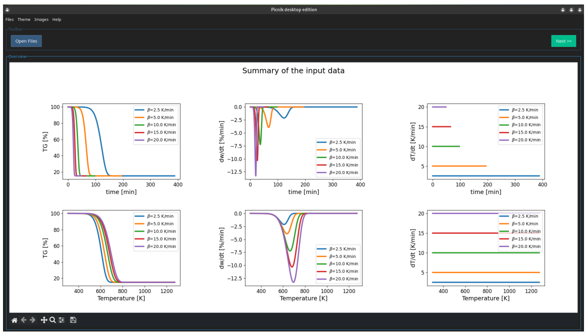
**WHAT YOU GET**

- Summary view displaying the six fundamental thermogram plots

`Action` In the File sub-menu of the *menu bar*, select Open file, or select Open file in the *top frame* choose dataset files using the system file chooser.



(a)



(b)

Figure 2: File selection interface and initial diagnostic plot overview in pICNIK. (a) Result of selecting `Open file` in the *top frame*. (b) Data visualization panel in pICNIK desktop edition showcasing the six primary diagnostic plots: mass percentage (TG), rate of mass loss (DTG), and temperature rate ($\mathrm{d}T/\mathrm{d}t$) *vs.* time (top panels) and *vs.* temperature (bottom panels) across multiple heating rates ($\beta$).

## 2.2 Individual Curve Visualizations

*Examine specific physical curves individually across time and temperature domains.*

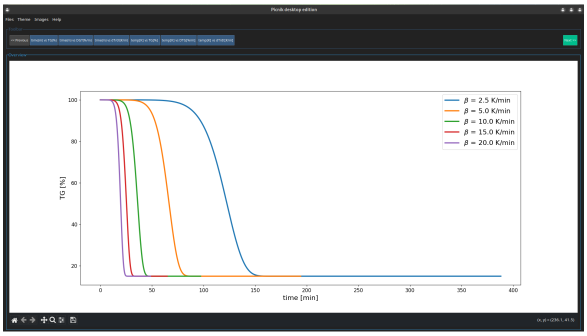
**WHAT YOU NEED**

- Data loaded in Step 1

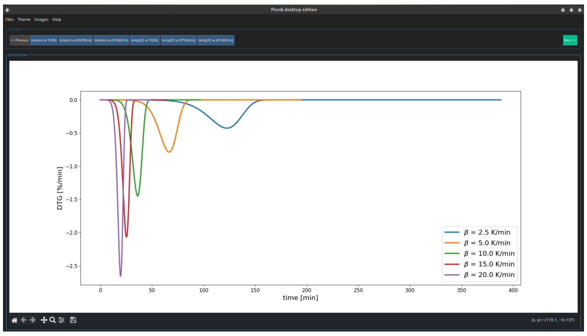
**WHAT YOU GET**

- Specialized views: TG *vs.* time, DTG *vs.* time $\mathrm{d}T/\mathrm{d}t$ *vs.* time, TG *vs.* temperature, DTG *vs.* temperature, and $\mathrm{d}T/\mathrm{d}t$ *vs.* temperature

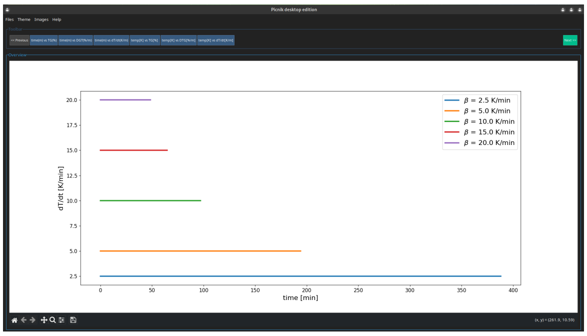
> **Action** Click corresponding *top frame* buttons to switch active bottom-frame visualization.



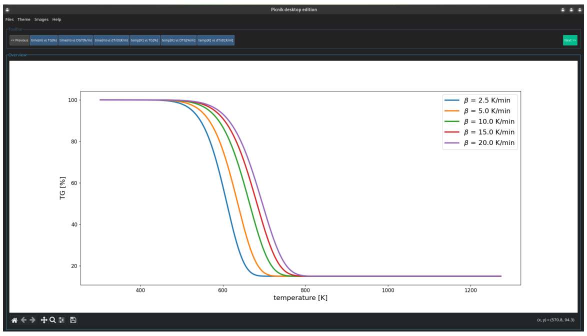
(a) TG *vs.* time



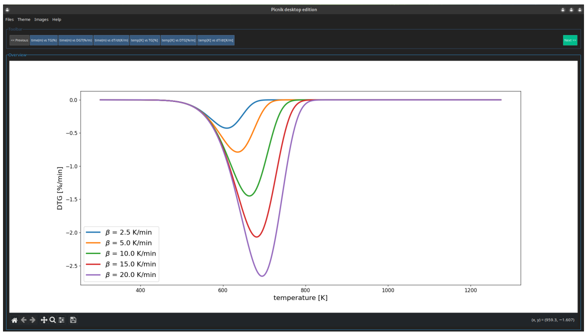
(b) DTG *vs.* time



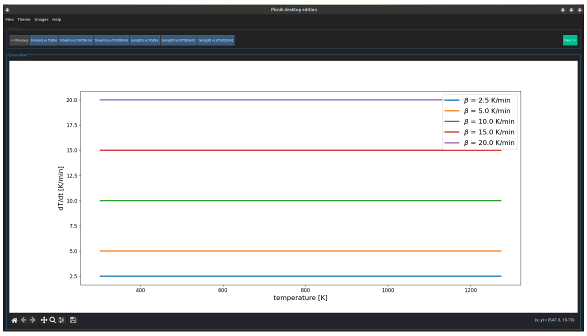
(c) d$T$/dt *vs.* time



(d) TG *vs.* temperature



(e) DTG *vs.* temperature



(f) d$T$/dt *vs.* temperature

Figure 3: Individual curve visualization.

## 2.3 Temperature Limits & Conversion Setup

*Define lower and upper temperature boundaries for each heating rate to calculate reaction conversion curves $\alpha(T)$.*

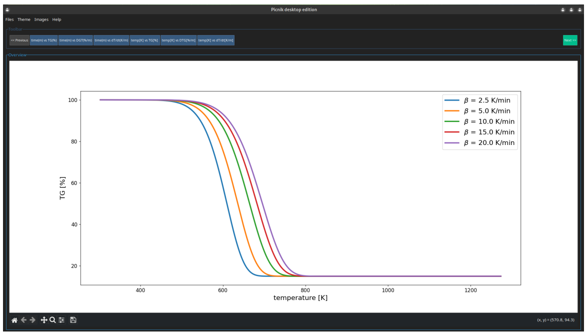
**WHAT YOU NEED**

- Inspection thermograms from previous step.

**WHAT YOU GET**

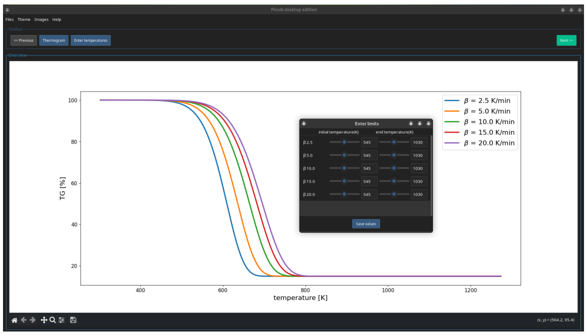
- Set temperature limits per curve and calculated conversion curves $\alpha(T)$.

`Action` `Click Thermogram button at` *`top frame`* `and enter lower and upper temperatures per` $\beta$`, save values and then, click Conversion button.`



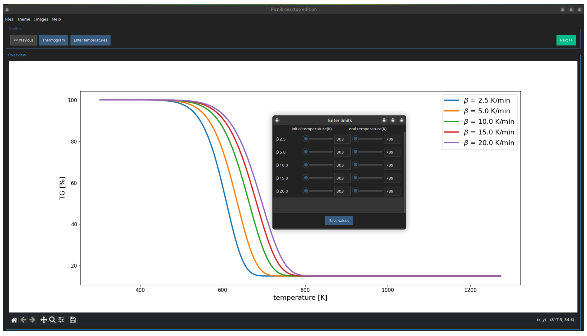


(a) TG *vs.* temperature





(b) Setting initial temperature limits





(c) Adjusting temperature boundaries



(d) Saving temperature limits

Figure 4: The setting of lower and upper temperatures of the thermograms at different heating rates ($\beta$).

## 2.4 Conversion Calculation

*The conversion is computed using the pICNIK library.*

**WHAT YOU NEED**

- Having defined the lower and upper temperature boundaries in the previous step.

**WHAT YOU GET**

- Conversion as a function of temperature

| Action | `Click Conversion button at` *`top frame`*`.` |
|---|---|

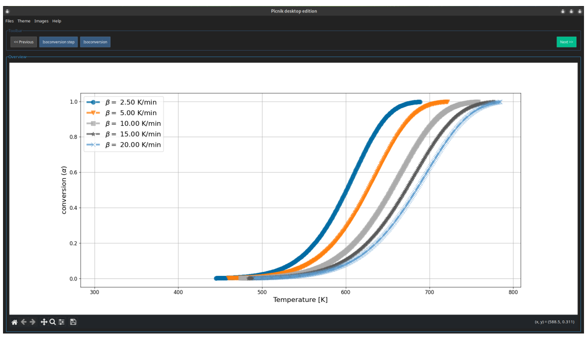




Figure 5: Conversion curves ($\alpha$ *vs.* temperature) for different heating rates ($\beta$).

## 2.5 Setting the conversion step

*The user enters the conversion step. The code uses these data to set the time steps for integration involves in activation energy calculations.*

WHAT YOU NEED

- Having completed the previous step.

WHAT YOU GET

- Conversion step value from user.

Action `Click Isoconversion step button at` *top frame*.

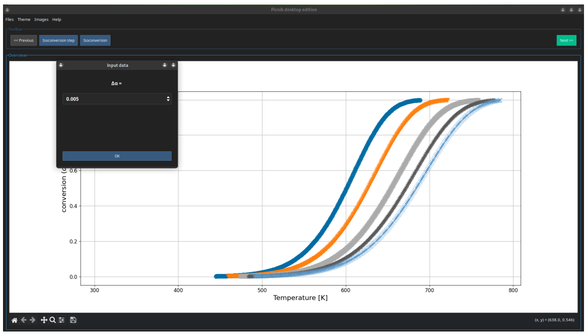


Figure 6: Dialog after clicking `Isoconversion step`.

## 2.6 Isoconversional Interpolation

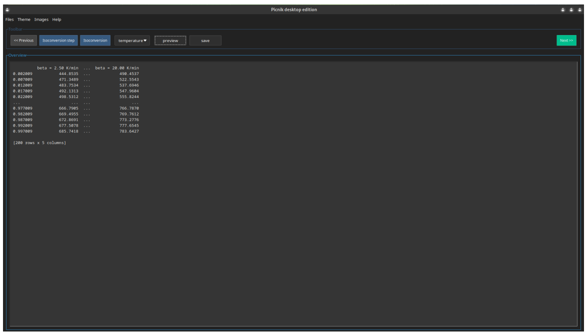
*The isoconversional tables are built: temperature, time, and conversion rate, interpolated to the same $\alpha$ values across all heating rates. This is the input required by the activation energy methods in step 2.7.*

WHAT YOU NEED

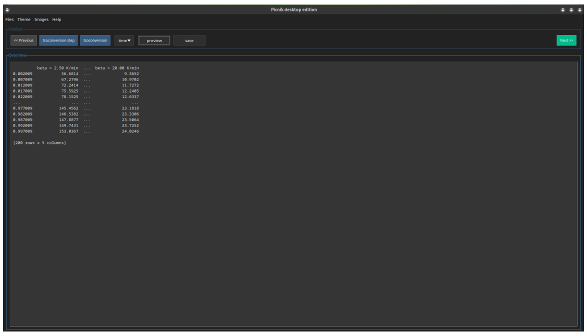
- Conversion increment $\Delta\alpha$
  `(default 0.02; adjustable between 0.001 and 0.1)`

WHAT YOU GET

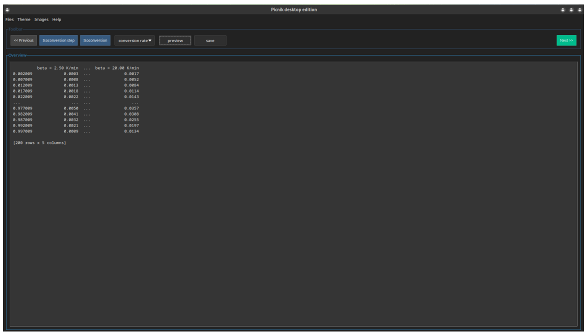
- 3 downloadable tables (temperature, time, conversion rate), indexed by $\alpha$



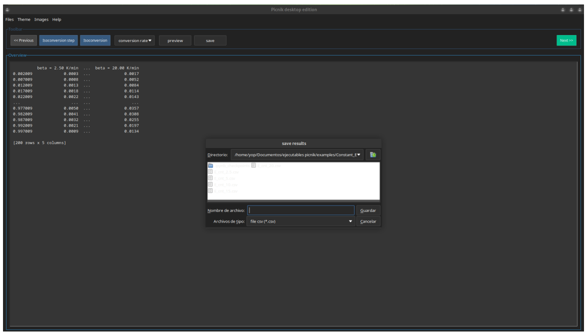
(a) Isoconversional table of Temperature generated from the $\Delta\alpha$ increment



(b) Isoconversional table of Time generated from the $\Delta\alpha$ increment



(c) Isoconversional table of Conversion rate generated from the $\Delta\alpha$ increment



(d) The image illustrates how to save the conversion rate table as an example

Figure 7: Isoconversional tables generated from the $\Delta\alpha$ increment.

## 2.7 Activation Energy Computation

*Calculate activation energy $E_\alpha$ using multiple model-free isoconversional methods.*

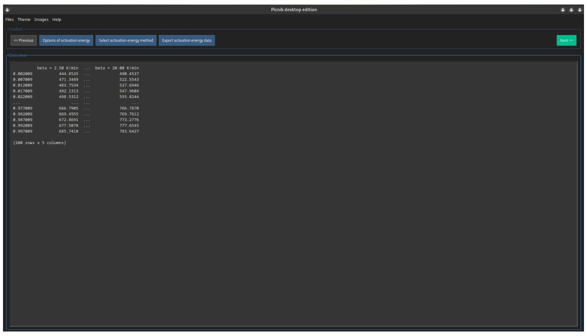
### WHAT YOU NEED

- Isoconversional tables from step 2.6
- Selection of methods: Friedman (Fr), Kissinger-Akahira-Sunose (KAS), Ozawa-Flynn-Wall (OFW), Vyazovkin (Vy), Advanced Vyazovkin (aVy)

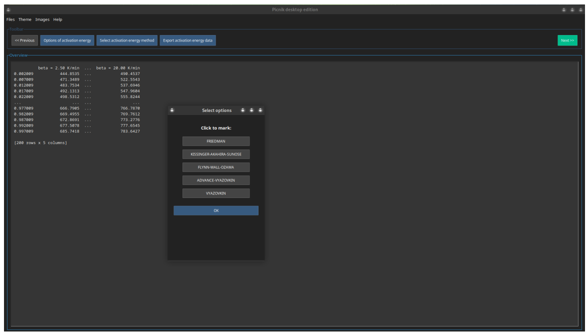
### WHAT YOU GET

- Interactive plot of $E_\alpha$ *vs.* $\alpha$ with error bars and data export options

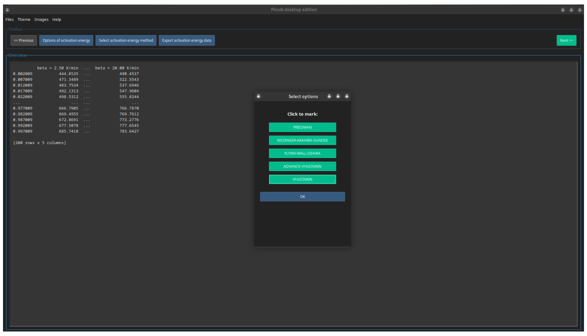
**Action** `Click Options of activation energy, mark desired methods, click OK, then select the active method via Select activation energy method.`



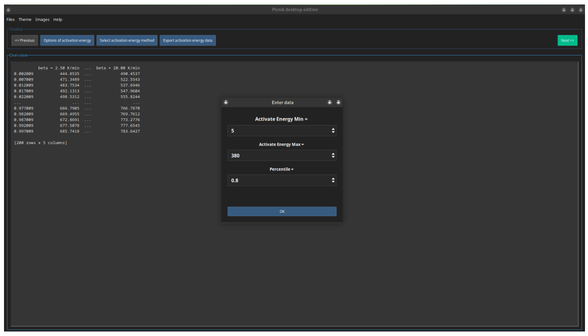
(a) Home screen



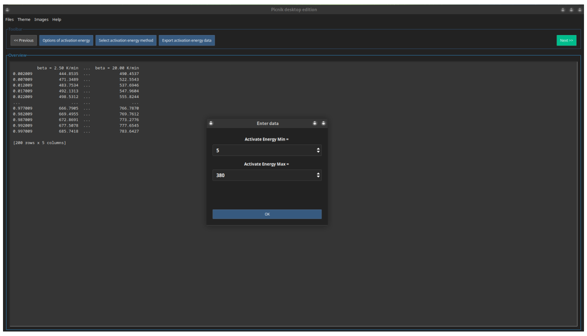
(b) Model selection



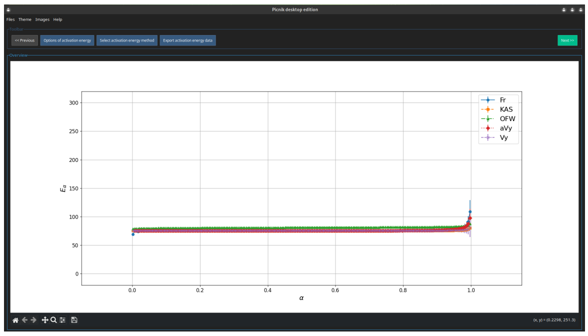
(c) Five selected models



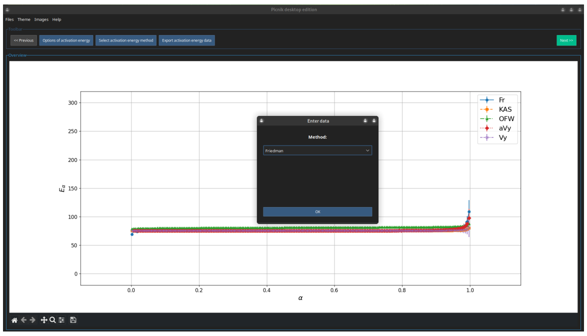
(d) Data entry for the aVy



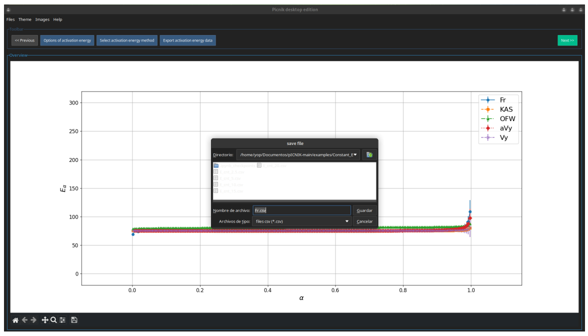
(e) Data entry for the Vy



(f) The different models are plotted.



(g) Model selection for calculating $E_\alpha$



(h) Save data for the selected model

Figure 8: Steps for activation energy computation

## 2.8 Compensation Effect Analysis

*Fit the kinetic compensation effect relation* $\ln(A) = a + bE$ *against candidate reaction models.*

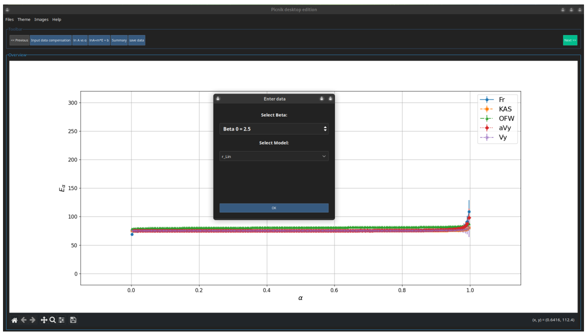
**WHAT YOU NEED**

- Computed activation energy $E_\alpha$
- Reference heating rate $\beta$ and model filtering criterion

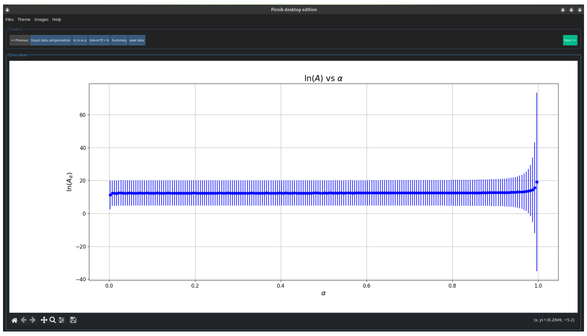
**WHAT YOU GET**

- Compensation parameters $a, b$, plot of $\ln A$ *vs.* $\alpha$, fit evaluation, and summary table

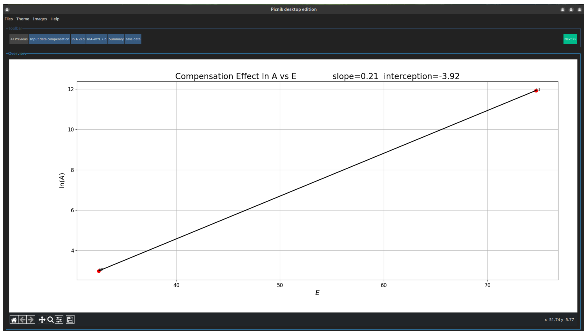
`Action` Click Input data compensation, select reference $\beta$ and model filter, then view $\ln(A)$ *vs.* $\alpha$, lnA=m*E+b, or summary.



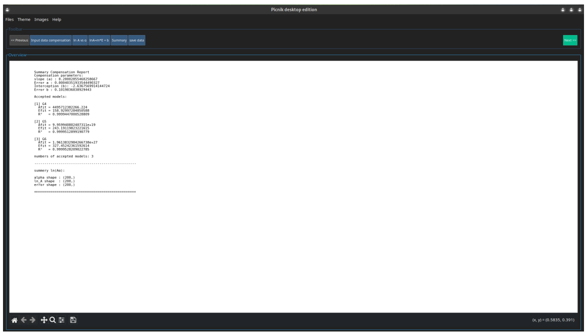
(a) model and $\beta$ selection



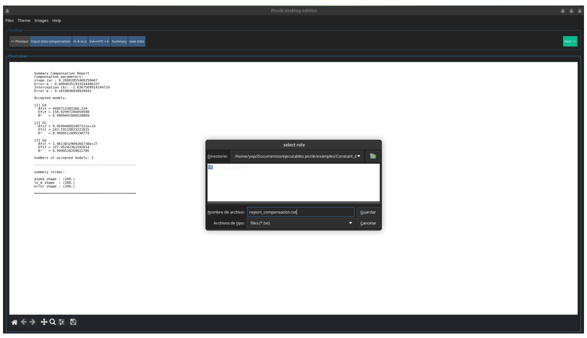

(b) Plot of $\ln(A)$ *vs.* $\alpha$




(c) Plot of $\ln(A)$ *vs.* $E$



(d) Summary compensation report



(e) Save compensation data

Figure 9: Steps for calculating the pre-exponential factor from the compensation effect

## 2.9 Model Reconstruction & Data Export

*Reconstruct the integral reaction model $g(\alpha)$ and save full kinetic parameters or reset session.*

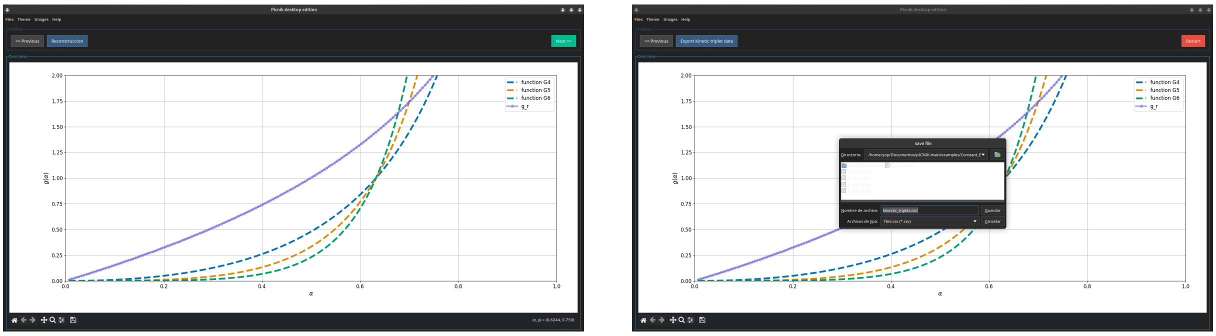
**WHAT YOU NEED**

- Activation energy $E_\alpha$ and pre-exponential factor $A_\alpha$

**WHAT YOU GET**

- Reconstructed $g(\alpha)$ curve compared against theoretical models and exported final dataset

`Action` `Click Reconstruction to view` $g(\alpha)$`, click save button to export kinetic data, or click Reset to restart analysis.`



(a) (b)

Figure 10: Numerically estimated $g(\alpha)$ and its theoretical values for different models (a) and export the kinetic triplet (b).

## 3 Technical Notes & deferred Validation

- **Deferred Validation System:** The application permits unrestricted navigation between workflow steps via top buttons. However, executing a calculation on a downstream screen without completing prerequisites triggers a safety message detailing the required missing input.

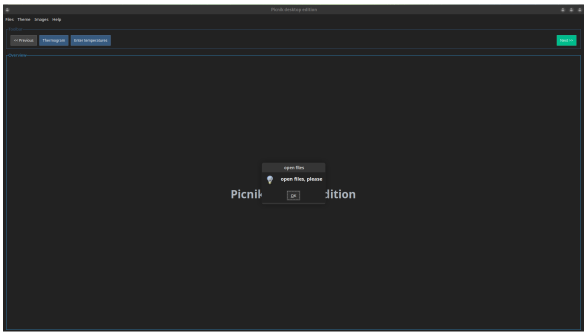
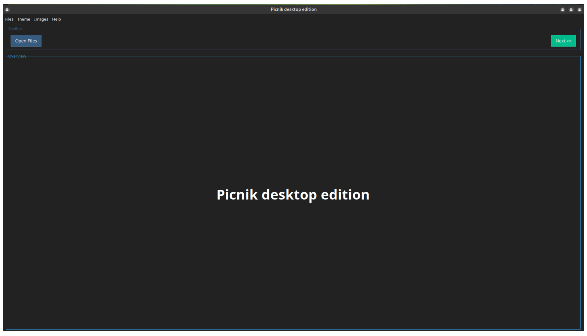
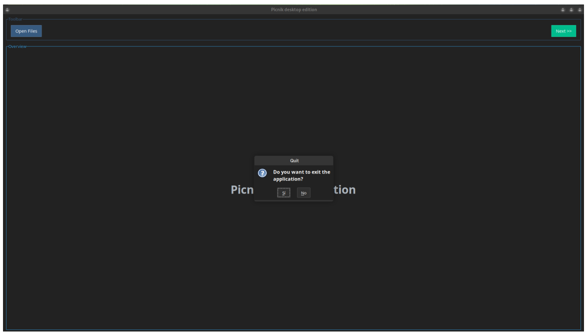


Figure 11: System validation

- **Data Export & File Exit:** Visualizations can be saved at any step using `Images` → `Save image` or the canvas toolbar. The application can be closed via `Files` → `Exit` or the window close button (`X`), both requiring user confirmation.




(a) Main interface




(b) Exit confirmation dialog

Figure 12: There are two ways to exit the application: the first is to select the “Exit” option from the “Files” menu, and the other is simply to click the close button at the top-right corner of the window. In both cases, you are asked to confirm that you want to exit, and once confirmed, the application closes.